Shaun, Troy, Scott

Development and regional water

How could new development affect an area's water supply and rivers through the year?

Project goal

Water available each season
Water used and returned
Town reserves + local rivers

Working demo

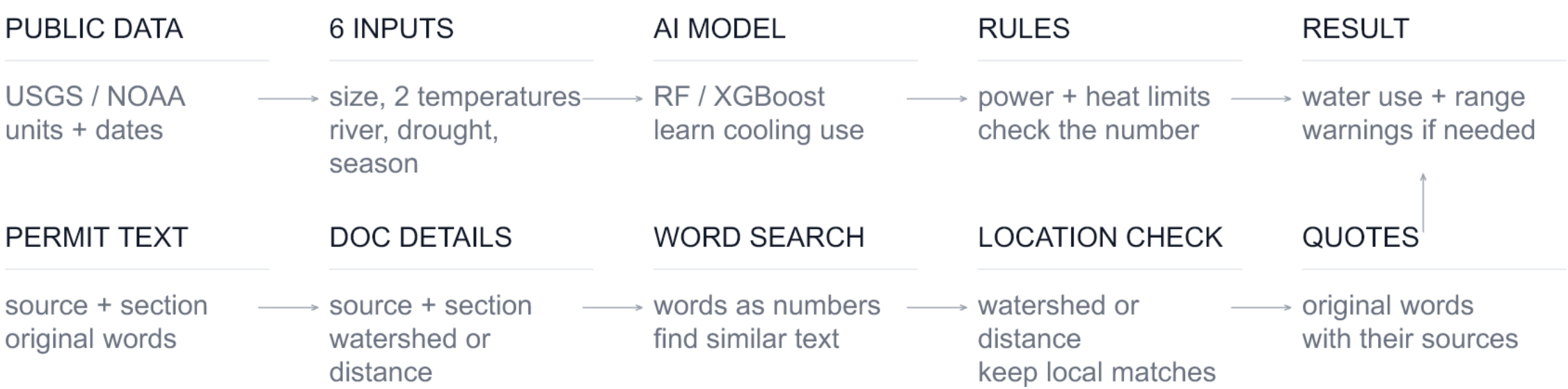
Weather data + cooling-water model
Rules check numbers + permit quotes
Full area assessment still to build

Cooling is one example. Users: town planners, researchers, water providers.

Current tests use generated values. Next: train and test on real measurements.

Current system: semantic map

Demo request: equipment size, cooling type, watershed code, month. Exact location is optional.

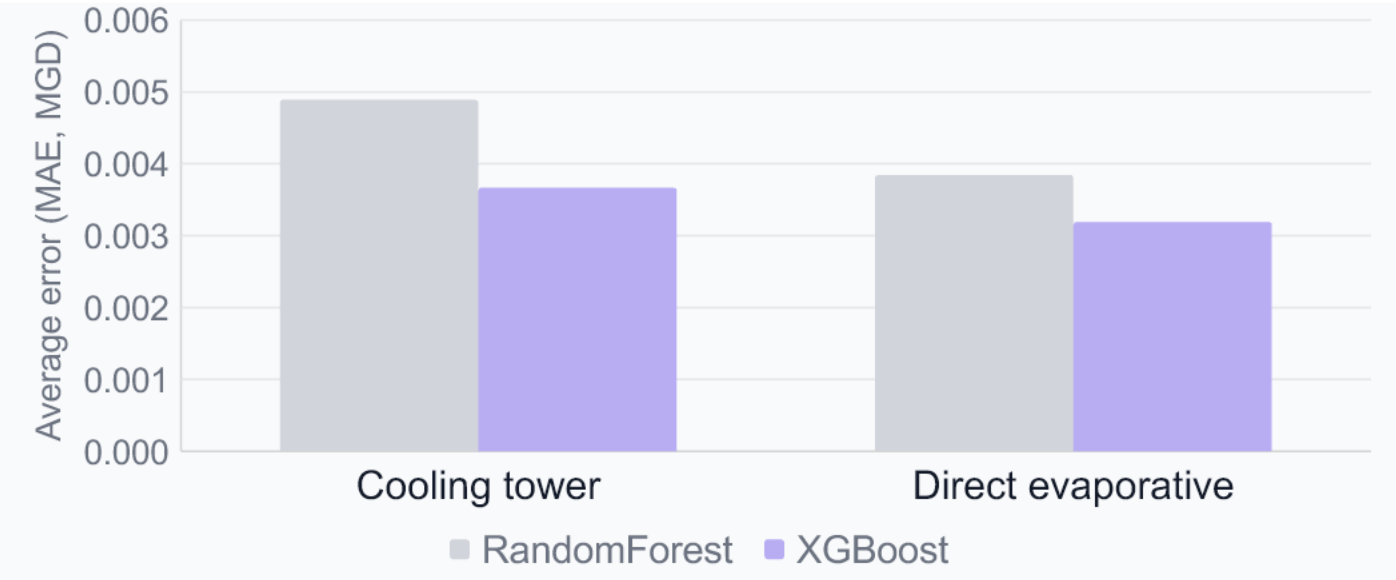


The AI estimates water use. Rules check it. Local permit quotes explain the context.

E3: forced wrong results stay within 0-0.446484 MGD at 40 MW. E4: all 4 location tests pass.

Evidence: E3/E4: constraint-failure-cases.json; retrieval-cases.json | Diagram shows current prototype

E1: Model comparison



XGBOOST: LOWER ERROR

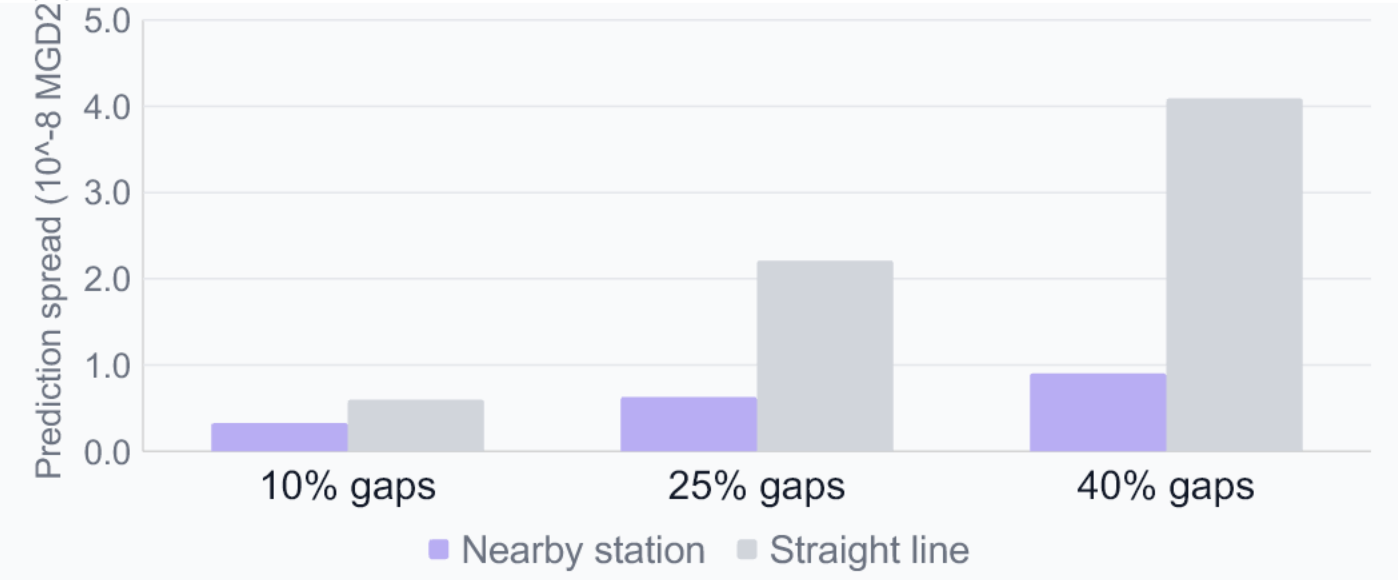
Pick using error on the middle date group. Keep RF to compare.

Next: train and test on real measurements.

Dates: train 60% / choose + set ranges 20% / test 20%
Dry air cooling uses zero water here. Its R2 fit score cannot be calculated.

Evidence: E1: regression-comparison.csv. 333 test rows per water-based cooling type. MGD = million US gallons/day.

E2: Missing drought readings



100%

All main tests passed

Change stayed within +/-0.05 MGD
50 repeats on 121 drought days

Failure: biased donors
only 60.3% within limits

Next: test bad neighbor readings and stations losing data together. The bad-reading test uses a simple formula.

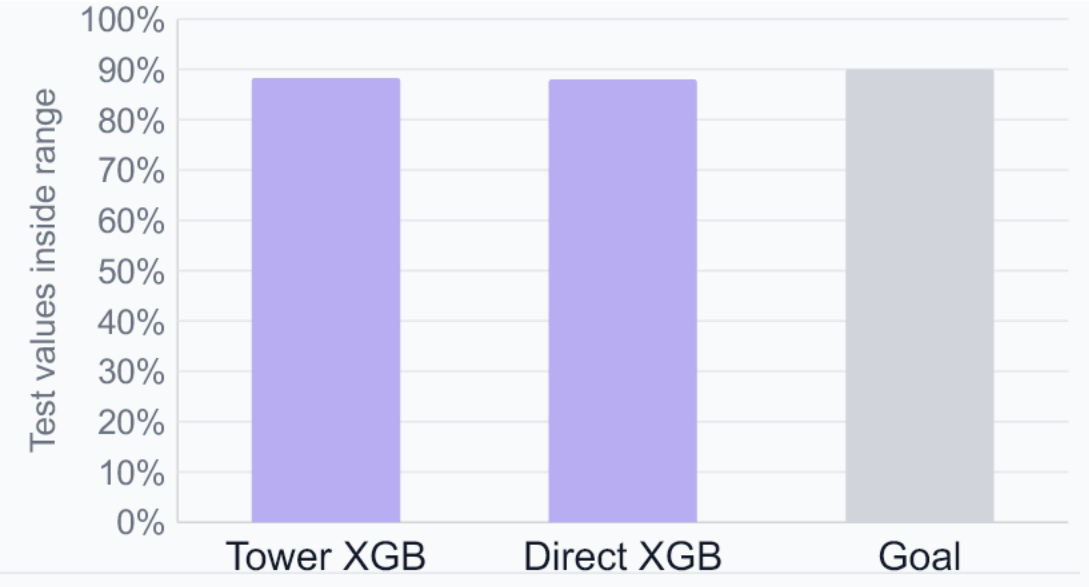
Evidence: E2: missingness-summary.csv, biased-donor-diagnostic.csv. Main test uses trained AI and made-up river data.

E5: Prediction ranges

88.3% / 88.0%

Coverage below the 90% goal

Tower fit score (R2) = 0.998
R2 measures fit. Coverage tests the range.



Improve: use separate data to choose models and set ranges.

A small test. We have not checked if the shortfall is more than chance. Real-world range tests are still needed.

Evidence: E5: interval-coverage.csv, benchmark.json. Made-up test values, 333 per model.

Decisions and next question

Adopt

E1: lower XGBoost error. E3/E4: checks pass.
Keep the models, limits, and local matches.

Modify

E2: bad neighbor readings. E5: below 90%.
Train on real data. Improve gaps/ranges + area totals.

Reject

E5: a fit score cannot promise a right answer.
Cooling use alone cannot show ecosystem health.

Defer

Need real area data before testing more AI methods.

Shaun: code + connect area data
Troy: data + source quotes
Scott: tests + results

Can training on real data predict effects on water reserves and rivers in a new area?

Evidence: E1-E5. Assigned roles: Shaun, Troy, Scott.